Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and Labor Markets — graduate level (Listas 3–4)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Consumption and saving

(Kurlat ch. 6; Romer ch. 8 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household lives two periods with $u(c_1) + \beta u(c_2)$, $u = c^{1-\sigma}/(1-\sigma)$, $\sigma = 2$, $\beta = 0.95$, and income (y_1, y_2). It earns $r_s = 2\%$ (real) on deposits but pays $r_b = 30\%$ (real) on consumer credit. For which values of y_2/y_1 does it neither borrow nor lend?

A. $[0.984,\ 1.111]$

B. $[0.969,\ 1.235]$

C. $[1.020,\ 1.300]$

D. $[0.950,\ 1.000]$

Answer: A

Step — kinked budget. The budget line has slope $-(1+r_s)$ to the right of the endowment (saving) and $-(1+r_b)$ to its left (borrowing). At the kink $c_t = y_t$.

Step — optimality at the kink. The household stays at the kink iff its MRS lies between the two gross rates: $1 + r_s \le \dfrac{u'(y_1)}{\beta u'(y_2)} = \dfrac{(y_2/y_1)^{\sigma}}{\beta} \le 1 + r_b$, i.e. $y_2/y_1 \in [(\beta(1+r_s))^{1/\sigma}, (\beta(1+r_b))^{1/\sigma}] = [0.984, 1.111]$. Option B is the log-utility ($\sigma = 1$) answer.

Ref: Kurlat (2020), §6.2, pp. 105–117, and Exercise 6.6 (kinked budget sets).

b) (3 points) Three periods, $r = 0$, wealth $W = 100$, log utility. The self at date t maximizes $\ln c_t + \beta \sum_{s>t} \ln c_s$ with $\beta = 1/2$ (present bias, no long-run discounting). Self 0 anticipates how self 1 will behave. What share of consumption *in every period* would self 0 give up to commit future selves to its own preferred plan?

A. 0%

B. 2.9%

C. 5.9%

D. 1.5%

Answer: B

Step — self 1. $\max \ln c_1 + \beta \ln c_2$ s.t. $c_1 + c_2 = W_1$: $c_1 = W_1/(1+\beta)$, $c_2 = \beta W_1/(1+\beta)$.

Step — self 0. $\max \ln c_0 + \beta[\ln c_1 + \ln c_2] = \ln c_0 + 2\beta \ln W_1 + \text{const}$: $c_0 = W/(1+2\beta) = 50$, so $W_1 = 50$, $c_1 = 33.3$, $c_2 = 16.7$. Self 0 would prefer $c_1 = c_2 = 25$.

Step — value of commitment. Gain $\beta[2\ln 25 - \ln 33.3 - \ln 16.7] = 0.0589$ utils. Cutting all three consumptions by a share x costs $(1+2\beta)\ln\frac{1}{1-x}$, so $x = 1 - e^{-0.0589/2} = 2.9\%$. Option A confuses "naive = sophisticated under log" (item c) with "no gain from commitment".

Ref: Romer (2012), §8.6 and problem 8.15, pp. 397–404; Kurlat (2020), §6.4, pp. 120–121.

c) (2 points, T or F) In (b), a sophisticated self 0 consumes less at date 0 than a naive self 0, who believes future selves will follow its plan, because it anticipates their overconsumption.

Answer: F

Step — naive. Believing $c_1 = c_2 = W_1/2$, the naive self solves $\max \ln c_0 + 2\beta \ln(W_1/2)$: $c_0 = W/(1+2\beta)$, the same.

Step — why. Under log utility, future consumption is proportional to W_1 whatever the rule, so the misbehavior changes continuation utility by a constant that does not depend on W_1 and leaves the margin untouched. With $\sigma \neq 1$ the two differ (the check script verifies that the sophisticate consumes less at $\sigma = 2$ and more at $\sigma = 1/2$); log is the knife edge.

Ref: Romer (2012), problem 8.15.

d) (2 points, T or F) A household with $y_1 = 0$ and $y_2 > 0$ that can borrow at r lowers c_1 when r rises, whatever $\sigma > 0$.

Answer: T

Step — closed form. From the Euler equation $c_2 = [\beta(1+r)]^{1/\sigma} c_1$ and the budget $c_1 + c_2/(1+r) = y_2/(1+r)$: $c_1 = \dfrac{y_2}{(1+r) + \beta^{1/\sigma}(1+r)^{1/\sigma}}$; the denominator rises with r for every $\sigma > 0$. Substitution, income (a borrower is poorer) and human-wealth effects all lower c_1.

Ref: Kurlat (2020), §6.2; Romer (2012), §8.4, pp. 380–384.

Question 2. (15 points) Discursive. Permanent income over an infinite horizon. A household maximizes $\sum_{t\geq 0} \beta^t u(c_t)$, $u = c^{1-\sigma}/(1-\sigma)$, subject to $c_t + a_{t+1} = y_t + (1+r)a_t$, a_0 given, and the no-Ponzi condition $\lim_{T\to\infty} a_{T+1}/(1+r)^T \geq 0$. Let $R \equiv 1 + r$ and $H_t \equiv \sum_{j\geq 0} y_{t+j} R^{-j}$.

a) (3 points) Derive the intertemporal budget constraint and the consumption function $c_0 = (1-\gamma)(Ra_0 + H_0)$, $\gamma \equiv \beta^{1/\sigma} R^{1/\sigma - 1}$. State the condition for a solution and show that with log utility the propensity to consume out of wealth does not depend on r.

Step — budget. Multiply the flow constraint by R^{-t}: $R^{-t}c_t + R^{-t}a_{t+1} = R^{-t}y_t + R^{-(t-1)}a_t$. Summing from 0 to T telescopes the assets:

$$\sum_{t=0}^{T} R^{-t} c_t + R^{-T} a_{T+1} = \sum_{t=0}^{T} R^{-t} y_t + R a_0.$$

At an optimum the no-Ponzi condition binds ($\lim R^{-T} a_{T+1} = 0$; a positive limit would be unspent wealth). Letting $T \to \infty$: $\sum_t R^{-t} c_t = Ra_0 + H_0$.

Step — Euler. Lagrangian with multiplier Λ on the budget: $\beta^t u'(c_t) = \Lambda R^{-t}$, so $u'(c_t) = \beta R u'(c_{t+1})$ and $c_t = c_0 (\beta R)^{t/\sigma}$.

Step — consumption function. Substituting: $c_0 \sum_t (\beta R)^{t/\sigma} R^{-t} = c_0 \sum_t \gamma^t = c_0/(1-\gamma)$, which needs $\gamma < 1$ (otherwise lifetime utility is unbounded and no optimum exists). Hence $c_0 = (1-\gamma)(Ra_0 + H_0)$.

Step — log case. $\sigma = 1$: $\gamma = \beta$, so the propensity to consume out of total wealth is $1 - \beta$, independent of r (income and substitution effects of r cancel; r still changes H_0).

Ref: Kurlat (2020), §6.3, eqs. (6.3.6)–(6.3.9), pp. 117–120; Romer (2012), §8.1.

Grading rubric.

- Telescoping to the intertemporal constraint, with the role of no-Ponzi: 1 pt.
- Euler equation: 1 pt.
- Consumption function and $\gamma < 1$: 0.5 pt.
- Log case: 0.5 pt.

b) (4 points) Let $\beta R = 1$. Show that $c_t = \frac{r}{R}(Ra_t + H_t)$ and prove Campbell's "saving for a rainy day" equation, $s_t \equiv ra_t + y_t - c_t = -\sum_{j\geq 1} R^{-j} \Delta y_{t+j}$.

Step — flat consumption. $\beta R = 1$ gives $\gamma = R^{-1}$, $1 - \gamma = r/R$, and c constant; applying (a) from date t: $c_t = \frac{r}{R}(Ra_t + H_t) = ra_t + \frac{r}{R} H_t$.

Step — saving. $s_t = ra_t + y_t - c_t = y_t - \frac{r}{R} H_t$.

Step — summation by parts. Expand the right-hand side of Campbell's equation:

$$-\sum_{j\geq 1} R^{-j}(y_{t+j} - y_{t+j-1}) = -\sum_{j\geq 1} R^{-j} y_{t+j} + R^{-1} \sum_{k\geq 0} R^{-k} y_{t+k} = -(H_t - y_t) + \frac{H_t}{R} = y_t - \frac{r}{R} H_t,$$

which is s_t (assuming $R^{-j}y_{t+j} \to 0$).

Economic intuition. Households save when they expect income to fall and dissave when they expect it to rise: saving is a forecast of income declines, and consumption reacts to news, not to current income.

Ref: Romer (2012), §8.2–8.3 (deterministic version).

Grading rubric.

- Flat consumption and the annuity formula: 1 pt.
- $s_t = y_t - \frac{r}{R}H_t$: 1 pt.
- Summation by parts, every line shown: 2 pts. Verifying on an example only: max. 1 pt.

c) (4 points) Brazil's 2019 pension reform, stylized. $r = 4\%$, $\beta R = 1$, $a_0 = 0$; earnings are 100 for $t = 0, \dots, 29$ and the pension is 60 forever from $t = 30$. At $t = 0$ a reform is announced that cuts the pension to 45. Compute Δc_0 with the consumption function and Δs_0 with Campbell's equation. Compare with the response to a one-off transfer and to a permanent raise of the same size, and give the limit of Δc_0 as $r \to 0$ (keeping $\beta R = 1$).

Step — wealth effect. $\Delta H_0 = -15 \sum_{t \geq 30} R^{-t} = -15R^{-30}\frac{R}{r}$, so $\Delta c_0 = \frac{r}{R}\Delta H_0 = -15R^{-30} = -15 \cdot 0.3083 = -4.62$.

Step — Campbell. Only Δy_{30} changes, by -15: $\Delta s_0 = -R^{-30}(-15) = +4.62 = -\Delta c_0$. Consistent.

Step — benchmarks. A one-off transfer X at $t = 0$: $\Delta c_0 = \frac{r}{R}X = 0.038X$. A permanent raise X: $\Delta H_0 = XR/r$, $\Delta c_0 = X$.

Step — limit. $-15R^{-30} \to -15$ as $r \to 0$: a fully patient household cuts consumption today by the whole future cut, because the infinite future weighs as much as the present.

Economic intuition. News about income 30 years ahead moves consumption today by 4.6% of current earnings; the pension reform raised private saving before any pension was paid.

Ref: Kurlat (2020), §6.3; Romer (2012), §8.2.

Grading rubric.

- ΔH_0 and Δc_0: 1.5 pts.
- Campbell check: 1 pt.
- Benchmarks: 1 pt. Treating the cut as transitory: max. 2 pts in the item.
- Limit: 0.5 pt.

d) (4 points) Now $a_{t+1} \geq 0$ for all t (no borrowing), log utility, $r = 4\%$, $\beta R = 1$, $a_0 = 0$; earnings are 50 for $t \leq 9$ and 100 from $t = 10$. Show the unconstrained plan is infeasible. Prove, with the Kuhn–Tucker conditions, that $c_t = y_t$ is optimal and find which multipliers are positive. Compute the welfare cost of the constraint (the share of the unconstrained consumption path the household would accept to give up) and the propensity to consume a small transfer received at $t = 0$ and at $t = 9$.

Step — infeasible. Unconstrained $c = \frac{r}{R}H_0 = 50 + 50R^{-10} = 83.78 > 50 = y_0$, so $a_1 < 0$.

Step — Lagrangian. $\mathcal{L} = \sum_t \beta^t \ln c_t + \sum_t \Lambda_t[y_t + Ra_t - c_t - a_{t+1}] + \sum_t \mu_{t+1}a_{t+1}$, $\mu \geq 0$, $\mu_{t+1}a_{t+1} = 0$. FOCs: $\beta^t/c_t = \Lambda_t$ and $-\Lambda_t + R\Lambda_{t+1} + \mu_{t+1} = 0$, so $\mu_{t+1} = \beta^t(\frac{1}{c_t} - \frac{1}{c_{t+1}})$ using $\beta R = 1$.

Step — verification. At $c_t = y_t$, $a_t = 0$: $\mu_{t+1} = 0$ for $t \leq 8$ and $t \geq 10$ (consumption flat); $\mu_{10} = \beta^9(\frac{1}{50} - \frac{1}{100}) > 0$. All multipliers are non-negative and complementary slackness holds; the objective is concave and the constraints linear, so the KKT point is optimal.

Step — welfare. $U^c = \sum_{t<10} \beta^t \ln 50 + \frac{\beta^{10}}{1-\beta} \ln 100$, $U^u = \frac{\ln 83.78}{1-\beta}$. The cost x solves $\frac{\ln(1-x)}{1-\beta} = U^c - U^u$: $x = 4.68\%$ of consumption forever.

Step — propensities. At $t = 9$ the household wants to borrow against $t = 10$ ($\mu_{10} > 0$): a small transfer is consumed at once, MPC $= 1$. At $t = 0$ the Euler equation holds with equality between dates $0, \dots, 9$, so the transfer is spread over those ten dates: $\Delta c_0 = X\frac{r/R}{1-R^{-10}} = 0.119X$.

Economic intuition. A household at zero wealth is not automatically hand-to-mouth at the margin: what matters is whether the constraint binds *next* period.

Ref: Kurlat (2020), §6.2 (borrowing constraints) and Exercise 6.5; Romer (2012), §8.6, pp. 393–395.

Grading rubric.

- Infeasibility: 0.5 pt.
- KKT conditions with signed multipliers: 1.5 pts. Asserting $c = y$ without the multipliers: max. 0.5 pt here.
- Welfare cost: 1 pt.
- Both propensities, with the timing argument: 1 pt.

Block 2 — Labor supply and search

(Kurlat ch. 7; Romer ch. 10 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) Let $u = \ln c - \psi \frac{h^{1+1/\varepsilon}}{1+1/\varepsilon}$ with $c = wh + \pi$, evaluated at $\pi = 0$, $\varepsilon = 1/2$. The Marshallian, Hicksian (compensated) and Frisch elasticities of hours with respect to w are:

A. $(0,\ 1/3,\ 1/2)$

B. $(1/2,\ 1/2,\ 1/2)$

C. $(0,\ 1/2,\ 1/3)$

D. $(-1/3,\ 0,\ 1/2)$

Answer: A

Step — FOC. $\psi h^{1/\varepsilon} = w/c$. Marshallian: at $\pi = 0$, $c = wh$ gives $\psi h^{1+1/\varepsilon} = 1$: hours do not depend on w.

Step — Frisch. Holding the marginal utility of wealth $1/c$ fixed, $h \propto w^{\varepsilon}$: elasticity $1/2$.

Step — Hicksian. Implicit differentiation of $\psi h^{\phi}(wh + \pi) = w$ ($\phi = 1/\varepsilon$) gives $\partial h/\partial \pi = -1/[w(1+\phi)]$ at $\pi = 0$. Slutsky: $\partial h^c/\partial w = \partial h/\partial w - h\,\partial h/\partial \pi$, so the compensated elasticity is $1/(1+\phi) = \varepsilon/(1+\varepsilon) = 1/3$. Frisch $\geq$ Hicksian $\geq$ Marshallian.

Ref: Kurlat (2020), §7.2, eq. (7.2.4), pp. 134–135, and footnote 8, p. 149.

b) (3 points) Prescott's calculation with $u = \ln c + \alpha \ln(1-h)$, $Y = K^{\vartheta}(hN)^{1-\vartheta}$, $\vartheta = 1/3$, a labor tax wedge τ rebated lump-sum. Calibrate α on the US ($h = 0.40$, $\tau = 0.40$, $c/y = 0.75$) and predict Brazil's hours ($\tau = 0.50$, $c/y = 0.80$):

A. 0.400

B. 0.357

C. 0.342

D. 0.333

Answer: C

Step — FOC. $\alpha c/(1-h) = (1-\tau)w$ and $w = (1-\vartheta)y/h$ give $\dfrac{h}{1-h} = \dfrac{(1-\tau)(1-\vartheta)}{\alpha\, c/y}$.

Step — calibration. US: $\frac{0.4}{0.6} = \frac{0.6 \cdot 2/3}{0.75\alpha} \Rightarrow \alpha = 0.8$. Brazil: $\frac{h}{1-h} = \frac{0.5 \cdot 2/3}{0.8 \cdot 0.8} = 0.521$, $h = 0.342$. Option B keeps $c/y = 0.75$.

Ref: Kurlat (2020), §7.3 and Exercise 7.5, pp. 137–140 and 148–149.

c) (2 points, T or F) With matching function $m = \mu U^{1-\alpha} V^{\alpha}$, the search equilibrium with bargained wages is efficient if and only if the workers' bargaining power equals α.

Answer: F

Step — Hosios. Efficiency requires the workers' power to equal the elasticity of matches with respect to *unemployment*, $1-\alpha$ (Question 4(c)).

Ref: Romer (2012), §10.7, pp. 497–498, and problem 10.17.

d) (2 points, T or F) In the steady-state search model with $w = (1-\phi)z + \phi(y + \kappa\theta)$, if unemployment income z and the vacancy cost κ are proportional to productivity y, a permanent rise in y leaves unemployment unchanged.

Answer: T

Step — homogeneity. The job-creation condition $(1-\phi)(y-z)q(\theta) = \kappa(r+s+\phi f(\theta))$ is homogeneous of degree one in (y, z, κ); with $z = \zeta y$, $\kappa = ky$, y cancels and θ, hence u, is unchanged.

Ref: Kurlat (2020), §7.5, p. 146 (Shimer's critique).

Question 4. (15 points) Discursive. The search-and-matching model in steady state (monthly). Labor force 1; unemployment u, vacancies v, tightness $\theta = v/u$; matches $m = \mu u^{1-\alpha} v^{\alpha}$; job-finding rate $f(\theta) = m/u = \mu\theta^{\alpha}$; vacancy-filling rate $q(\theta) = m/v = \mu\theta^{\alpha-1}$; separations at rate s. A filled job produces y and pays w; a vacancy costs κ per month; entry is free. Take the wage rule $w = (1-\phi)z + \phi(y + \kappa\theta)$ as given (it comes from Nash bargaining, ϕ = workers' power). Use one fact: a flow x that lasts until a Poisson event with rate ϱ is worth $x/(r+\varrho)$. Numbers: $r = 0.004$, $s = 0.03$, $\mu = 0.3$, $\alpha = 0.5$, $\phi = 0.5$, $y = 1$, $z = 0.7$.

a) (3 points) Derive the Beveridge curve from flow balance and prove that, in the (u, v) plane, it is downward sloping and convex.

Step — flows. $\dot{u} = s(1-u) - f(\theta)u$. Steady state: $u = \dfrac{s}{s + f(\theta)}$.

Step — in (u, v). $s(1-u) = \mu u^{1-\alpha} v^{\alpha} \Rightarrow v = C(1-u)^{1/\alpha} u^{-(1-\alpha)/\alpha}$, $C = (s/\mu)^{1/\alpha}$.

Step — shape. Both factors are positive, decreasing and convex in u ($1/\alpha \geq 1$). For a product fg of such functions $(fg)' = f'g + fg' < 0$ and $(fg)'' = f''g + 2f'g' + fg'' > 0$, since $f'g' > 0$.

Ref: Kurlat (2020), §7.1 and §7.5, eq. (7.5.2); Romer (2012), §10.6.

Grading rubric.

- Steady-state flow balance: 1 pt.
- Curve in (u, v): 1 pt.
- Slope and convexity proved: 1 pt.

b) (4 points) Derive the job-creation condition from free entry, combine it with the wage rule, and prove that tightness θ^* exists and is unique. Calibrate κ so that $\theta^* = 1$ and compute u^* and v^*. Sign $d\theta^*/dy$, $d\theta^*/dz$, $d\theta^*/d\phi$ and $d\theta^*/d\mu$.

Step — filled job. A flow $y - w$ until separation: $J = \dfrac{y-w}{r+s}$.

Step — vacancy. A flow $-\kappa$ until filled (rate q), then worth J: its value is $\dfrac{-\kappa + qJ}{r+q}$. Free entry sets it to zero: $\kappa/q(\theta) = J$ (expected recruiting cost = value of a filled job).

Step — job creation. With the wage rule, $y - w = (1-\phi)(y-z) - \phi\kappa\theta$, so

$$(1-\phi)(y-z) = \kappa\Big[\frac{r+s}{q(\theta)} + \phi\theta\Big] \iff (1-\phi)(y-z)q(\theta) = \kappa(r+s+\phi f(\theta)).$$

Step — existence and uniqueness. The bracket is continuous and strictly increasing in θ ($1/q = \theta^{1-\alpha}/\mu$), equals 0 at $\theta = 0$ and tends to ∞: if $y > z$ there is exactly one $\theta^* > 0$.

Step — calibration. At $\theta = 1$: $\kappa = \dfrac{(1-\phi)(y-z)\mu}{r+s+\phi\mu} = \dfrac{0.045}{0.184} = 0.2446$; $f = 0.3$, $u^* = 0.03/0.33 = 9.09\%$, $v^* = \theta^* u^* = 9.09\%$.

Step — comparative statics. Write $G(\theta) \equiv \kappa[(r+s)\theta^{1-\alpha}/\mu + \phi\theta] - (1-\phi)(y-z) = 0$, $G_\theta > 0$. $G_y < 0 \Rightarrow d\theta^*/dy > 0$; $G_z > 0 \Rightarrow d\theta^*/dz < 0$; $G_\phi = \kappa\theta + y - z > 0 \Rightarrow d\theta^*/d\phi < 0$; $G_\mu < 0 \Rightarrow d\theta^*/d\mu > 0$ (implicit function theorem: $d\theta/dx = -G_x/G_\theta$).

Ref: Romer (2012), §10.6, pp. 486–492 (present values instead of asset equations); Kurlat (2020), §7.5, eq. (7.5.1).

Grading rubric.

- J and the free-entry condition as present values: 1 pt.
- Job-creation condition with the wage: 1 pt.
- Existence and uniqueness: 1 pt. Graph only: 0.5 pt.
- Calibration and the four signs: 1 pt.

c) (4 points) Efficiency with $r \to 0$. A planner chooses θ to maximize steady-state net output $y(1-u) + zu - \kappa v$ subject to the Beveridge curve. Derive its condition and prove that the market θ coincides with it if and only if $\phi = 1 - \alpha$. With κ from (b), compute efficient θ and u, and the market values when $\phi = 0.7$. Explain the externalities.

Step — objective. $W(\theta) = y - (y-z)u(\theta) - \kappa\theta u(\theta)$, $u = \frac{s}{s+f}$, $u'(\theta) = -\frac{u\, f'(\theta)}{s+f}$, $f' = \alpha q$.

Step — planner FOC. $W' = -(y - z + \kappa\theta)u' - \kappa u = 0 \Rightarrow (y - z + \kappa\theta)\frac{\alpha q}{s+f} = \kappa \Rightarrow \alpha(y - z + \kappa\theta) = \kappa\frac{s}{q} + \kappa\theta$, i.e.

$$\alpha(y - z) = \kappa\frac{s}{q(\theta)} + (1-\alpha)\kappa\theta.$$

Step — market at $r = 0$. From (b): $(1-\phi)(y-z) = \kappa\frac{s}{q(\theta)} + \phi\kappa\theta$. Both right-hand sides are increasing in θ, so the two solutions coincide for all parameters iff $1 - \phi = \alpha$ and $\phi = 1 - \alpha$: Hosios.

Step — numbers. Efficient: $\theta = 1.024$, $u = 8.99\%$ (the market with $\phi = 0.5$ reaches it). With $\phi = 0.7$: $\theta = 0.432$, $u = 13.21\%$.

Step — externalities. An extra vacancy raises f (helps the unemployed) and lowers q (congests other firms). Entry is efficient when the firm's share of the surplus, $1 - \phi$, equals its contribution to matching, α.

Ref: Romer (2012), §10.7, pp. 497–498, and problem 10.17, p. 512.

Grading rubric.

- Planner objective and FOC: 1.5 pts.
- Comparison and the Hosios condition: 1.5 pts. Stating $\phi = 1 - \alpha$ without the comparison: max. 1 pt in the item.
- Numbers and externalities: 1 pt.

d) (4 points) With $\alpha = 1/2$ the job-creation condition is a quadratic in $x = \sqrt{\theta}$: solve it. Compute θ^* and u^* after (i) mismatch lowers μ to 0.225 and (ii) more generous unemployment insurance raises z to 0.8. Derive the elasticity of θ^* with respect to y at the calibrated point, and of u^*. What does the formula say about Shimer's puzzle?

Step — quadratic. Multiply $(1-\phi)(y-z)\mu x^{-1} = \kappa(r + s + \phi\mu x)$ by x: $\kappa\phi\mu x^2 + \kappa(r+s)x - (1-\phi)(y-z)\mu = 0$, with positive root $x = \dfrac{-\kappa(r+s) + \sqrt{\kappa^2(r+s)^2 + 4\kappa\phi\mu^2(1-\phi)(y-z)}}{2\kappa\phi\mu}$.

Step — shocks. (i) $\mu = 0.225$: $\theta^* = 0.935$, $u^* = 12.12\%$; both the Beveridge curve shifts out and θ falls. (ii) $z = 0.8$: $\theta^* = 0.637$, $u^* = 11.14\%$.

Step — elasticity. Implicit differentiation of the job-creation condition at $\theta = 1$, with κ from (b):

$$\frac{d\ln\theta^*}{d\ln y} = \frac{y}{y-z}\cdot\frac{1}{(1-\alpha) + \alpha\phi f/(r+s+\phi f)} = 3.33\cdot\frac{1}{0.908} = 3.67,$$

and $\frac{d\ln u^*}{d\ln\theta} = -\alpha(1-u^*) = -0.455$, so $\frac{d\ln u^*}{d\ln y} = -1.67$.

Step — Shimer. The response is governed by $y/(y-z)$: with a large surplus (z far below y) the model generates little volatility in θ and u; matching the data requires a small surplus (z close to y) or rigid wages.

Ref: Kurlat (2020), §7.5, p. 146; Romer (2012), problems 10.12–10.15.

Grading rubric.

- Quadratic and its root: 1 pt.
- Both shocks: 1 pt.
- Elasticity derived by implicit differentiation: 1.5 pts.
- Shimer reading: 0.5 pt.